

# Decoding the Absence: Exploring the Influence of Negative Prompts in Stable Diffusion

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## Abstract

*Stable Diffusion models have the ability to generate photorealistic images from text prompts. When negative prompts are included in the image generation process using stable Diffusion, it provides a unique approach to guide the image generation process by specifying what the user does not want to see in generated images. However, the impact of incorporating negative prompts on the image-generation process using both positive and negative prompts is not well understood. Negative prompts do not always remove the object that they describe, and they add content as much as they subtract. In this work we showcase our efforts to understand negative prompts from the perspective of object removal tools: how effective is it at "removing" objects from a image strictly with image generation not inpainting, and from the point of view of "if not this then what?" can we find and decode a **fused prompt** which can by itself produce the same image as with the negative prompt while remaining interpretable.*

## 1. Introduction

Image generation models are something special because they are very much "you get what you see" in so far as a qualitative view of their outputs is a common initial evaluation approach. Diffusion models mark the beginning of a new leap in image generation performance. We have had ViTs, single-step diffusion, and even NERF improvements that are similar enough in quality to be thought of as the same class of model. However, as with many cutting-edge technologies, their operation often encompasses intricate behaviors that remain elusive to comprehension. Notably in diffusion, the unexpected outputs arising from the complex interaction of positive and negative prompts confirm the underlying complexities yet to be unraveled.

The role of prompts in guiding the image generation process is pivotal to diffusion. While positive prompts guide

the model toward desired characteristics, negative prompts serve to deter specific outcomes. However, the actual semantic transformations these prompts induce remain an area of intriguing uncertainty. Given this backdrop, the goal of this work is to show our explorations into how the negative prompt is "semantically transforming" images. When we say semantically transformed we mean a transformation of meaning or object in the image rather than simply a pixel color shift.

We show two explorations of this approach: first, we explore "if not this then what?" if we use a negative prompt to "remove" we want to know what is added in its place. We investigate this by defining a process of approximating a **fused prompt** 3.1, a prompt such that if given to stable diffusion will result in the same image produced with a given positive and negative prompt. We do this through a discrete optimization process described in 3.2. Second, to investigate "what is removed?", we construct a dataset of diffusion latents and distill it to images and latents with known objects present. Then we perform CLIPSeg segmentation over the same positive prompt image trying to locate the object, but with a provided negative prompt which is the name of that object. See this work here 4.3.

## 2. Background

Diffusion models, as a class of generative models, have established a new frontier in image generation by learning to iteratively add and subtract noise, often Gaussian, according to a specific variance schedule [1, 5]. This process, adhering to a Markov chain structure, allows for the generation of images from noise by progressively removing the predicted noise, a technique that has been recognized for producing superior results in text-conditioned image generation [13]. Central to these models is the way the noise distribution is conditioned on text, using encoded text tokens, often with CLIP, and integrating these into cross-attention mechanisms [12, 19]. This interplay is fine-tuned by the "guidance-scale" hyper-parameter, a critical component in

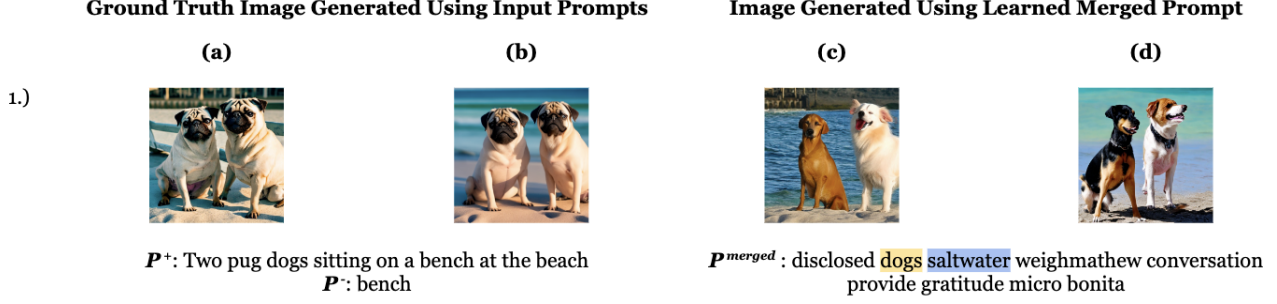


Figure 1. Diffusion Token Optimzation example showing original prompt, optimized fused prompt, and semantically similar images

the classifier-free guidance framework [4]. While the focus has traditionally been on positive prompts, directing the model towards specific outcomes, the introduction of negative prompts represents a significant leap forward [13].

Negative prompts invert the conventional model guidance, replacing "unconditional noise" with noise conditioned on additional negative text inputs. This innovative approach qualitatively alters the generated image, leading to the semantic omission or transformation of elements specified in the negative prompts. This development opens new avenues for controlling and refining the image generation process, enabling more precise and diverse outputs.

However, despite their advanced capabilities, diffusion models, particularly those employing structures like Unet or other cross-attention mechanisms, are faced with a trade-off between their ability to learn complex representations (expressiveness) and the clarity of their operational mechanisms (explainability) [13]. While they excel at creating high-fidelity images closely adhering to the provided prompts, a comprehensive understanding of their behavior in response to various, especially complex or contradictory, inputs remains elusive. Decoding the Absence aims to elucidate this aspect of diffusion models, specifically focusing on the role and impact of negative prompts. We investigate how these prompts influence the image generation process beyond mere object removal, examining their contribution to the overall semantic content and complexity of the generated images. This exploration is key in enlarging our understanding of the capabilities and limitations of stable diffusion models, particularly in the context of intricate text-conditioned image synthesis.

### 3. Methodology

#### 3.1. Fused Prompt Optimization

In this section, we explore a novel optimization process aimed at acquiring a fused prompt that replicates the noise generated through classifier-free guidance. Our goal is to comprehend how substituting a negative prompt for uncon-

ditioned noise in classifier-free guidance influences stable diffusion image generation. Figure 2 outlines our optimization process, illustrating how we leverage the original pair of positive and negative prompts to learn a fused prompt. This fused prompt is designed to generate an image with a comparable intent to the one produced using the original pair of prompts.

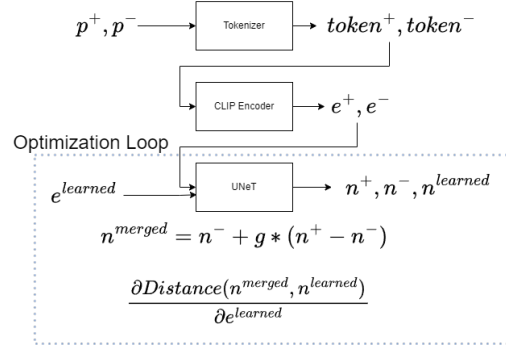


Figure 2. Optimization Loop

In Figure 2, we illustrate the optimization process of  $e^{learned}$ , ensuring that  $n^{learned}$  closely aligns with  $n^{merged}$ . What makes this method intriguing is its potential to yield distinct  $n^{learned}$  values for each timestep, diverging from the behavior of  $n^{merged}$ . Here,  $p^+$  and  $p^-$  represent the positive and negative prompts, while  $token^+$  and  $token^-$  denote their tokenized equivalents. The positive and negative text embeddings,  $e^+$  and  $e^-$ , are generated using CLIP Text Encoder, as we have been using Stable Diffusion 2.1.

$$\tilde{\epsilon}(Z, w, y) = \epsilon(Z) + w * (\epsilon(Z, y) - \epsilon(Z)) \quad (1)$$

The classifier-free guidance equation (see Equation 1 [4]) can be understood as a sequence of Euclidean transforms applied to the latents generated by the UNeT. The conditional and the unconditional distributions are modelled by the same UNeT. A crucial parameter in this equa-

tion is  $w$ , which serves as a control mechanism for the diversity of the generated samples. When  $w = 1$ , the equation essentially performs only unconditional sampling.

In essence, our objective of finding a merged prompt is akin to stating:

$$UNeT(x) + UNeT(y) = UNeT(z)$$

Our goal is to determine the value of  $z$  when given  $x$  and  $y$ , or, in the context of our optimization, to identify  $e^{learned}$  such that  $n^{learned}$  equals  $n^{merged}$  at every timestep  $T$  during our diffusion process. Within our optimization loop, we aim to align  $n^{learned}$  closely with  $n^{merged}$ , necessitating three passes through UNeT in each iteration—one for positive, one for negative, and one for learned embeddings. To achieve this, we must unfreeze the UNeT layers, allowing gradients to pass through and modify the learned embeddings based on the loss. We experimented with different loss functions to compare noises and latents, and our observations indicate that Mean Squared Error (MSE) yields the most effective results.

### 3.1.1 Time Configurations

We’ve explored two distinct strategies—Semi-global Optimization and Randomized Optimization—to obtain the learned text embedding ( $e^{learned}$ ). In Semi-global Optimization, our focus is on accumulating gradients across multiple timesteps and applying them iteratively, specifically once every third step. The objective is to fine-tune the learned embeddings ( $e^{learned}$ ). On the other hand, Randomized Optimization adopts a strategy where we randomly select a timestamp during each epoch for backpropagation, inspired by training techniques used in stable diffusion models.

Given the multi-step nature of diffusion, we opted for these varied strategies to accommodate different timesteps. Our experimentation revealed that the Semi-Global Optimization strategy outperformed others, successfully generating images with a similar intent as the ground truth. We attribute this success to the observation that the scheduler step depends on the previous latents.

Despite the effectiveness of these strategies, it’s noteworthy that neither approach allowed us to generate the exact same image using the fused prompt. This limitation highlights a constraint within our optimization approach.

In each of these strategies, we began by initializing the learned embedding, denoted as  $e^{learned}$ , with  $e^+$ , and employed the Adam optimizer [6]. Our primary objective was to align  $n^{learned}$  with  $n^{merged}$ , treating it as a task similar to assessing the similarity between two samples drawn from a probability distribution. To quantify this similarity, we explored various methods or functions suited for comparing two samples originating from the distribution. These

included Earth Movers’ Distance [14], KL Divergence [8], LPIPS [21], Mean Square Error (MSE), or a combination of these functions. We assigned varying weights to scale them in our optimization process. Through experimentation, we found that Mean Square Error (MSE) yielded the most effective results. Figure 5 displays some examples of images generated using the learned embedding  $e^{learned}$ .

## 3.2. Discrete Token Optimization

To optimize the model with respect to tokens and directly obtain  $p^{merged}$ , we utilized the PEZ algorithm [20]. For direct optimization with tokens, we represent the learned embedding as  $e^{learned} = [e_1, e_2, \dots, e_M]$ , where  $e^{learned} \in \mathbb{R}^{M \times d}$ , with  $M$  as the number of **tokens** and  $d$  as the dimension of the embedding. We employed a projection function,  $Proj_E$ , which maps individual embedding vectors  $e_i$  from  $e^{learned}$  to their nearest neighbors in the embedding matrix  $E^{|V| \times d}$ , where  $|V|$  is the vocabulary size of the tokenizer (refer Figure 3).

Therefore, after applying this projection function, our resultant projected embedding  $(e^{learned})' = Proj_E(e^{learned})$ . During each forward pass, we first project the current learned embedding  $e^{learned}$  onto its nearest neighbor  $(e^{learned})'$  before calculating the gradient. Subsequently, using the gradient of the discrete vectors  $(e^{learned})'$ , we update the continuous embedding  $e^{learned}$ .

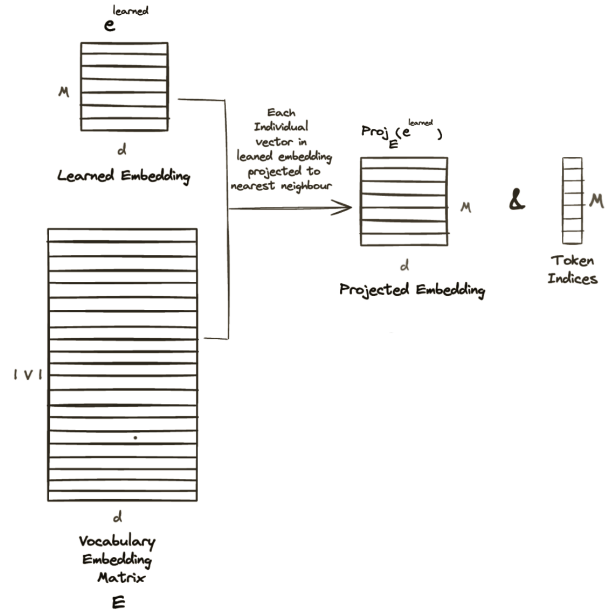


Figure 3. Discrete Token Optimization

## 4. Results

### 4.1. Fused Prompt Optimization Results

The results presented in both Figure 4 and Figure 5 are based on the experiments, where we employed the semi-global optimization strategy and utilized Mean Squared Error (MSE) as the loss function for learning the fused prompt.

In Figure 4, we present two sets of images: the ground truth and those generated using the fused prompt. The ground truth images are generated using the original positive and negative prompt pairs through the default stable diffusion pipeline. In contrast, the images generated with the fused prompt result from passing the learned embedding ( $e^{learned}$ ) through a customized stable diffusion pipeline. This pipeline conditions the image generation on  $e^{learned}$  and unconditional embeddings/sampling, mirroring the default behavior of classifier-free guidance. Each column in Figure 4 represents a distinct strategy employed to learn the fused prompt. These variations showcase different approaches taken in the optimization process.

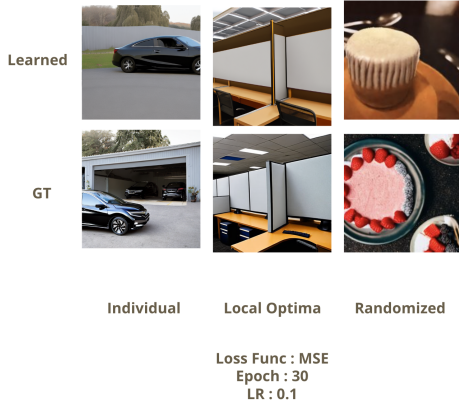


Figure 4. Prompt Fusion Examples

In Figure 5, additional sets of images generated using the fused prompt are displayed alongside their comparisons with the original ground truth images. The figure also presents the pairs of negative and positive prompts that were utilized to generate these ground truth images. Notably, the images generated with the fused prompts appear to maintain a similar intent to the original ground truth images. This observation underscores the effectiveness of the fused prompt in capturing and preserving the intended content during the image generation process.

### 4.2. DTO Results

In Figure 6, we showcase the resulting image generated through our learned **merged prompt** ( $P^{merged}$ ). The results reveal that images generated from  $P^{merged}$  exhibit semantic similarities with the ground truth image generated



Figure 5. Images generated using Fused Prompt Optimization

using both the Positive ( $P^+$ ) and Negative ( $P^-$ ) prompts upon visual inspection. For instance, the images generated using the merged prompt ( $P^{merged}$ ) in the first row of Figure 6 convey similar semantic information as those generated using the positive and negative prompts. In both images, a beach serves as the background, and two dogs are standing on the sand. While the dogs differ between the ground truth image (pug dogs) and the generated image (two random dogs), both images convey the same semantic message.

Moreover, the learned **merged prompt** ( $P^{merged}$ ) comprises both real words and gibberish text. However, the valid words within the prompts contribute significantly to conveying information about the image. In the first row, words such as **dogs** and **saltwater** emerge as primary elements in the ground truth image generated using both positive and negative prompts. Additionally, utilizing our  $P^{merged}$  allows us to generate a diverse set of images by employing different random seeds. These generated images maintain semantic similarity with those produced using the Positive and Negative Prompts.

### 4.3. Diffusion Explorative Dataset

Investigating how objects are "removed" by negative prompts is non-trivial because we are not performing inpainting and thus provide only a randomly seeded initial without knowledge of what it will produce. To remedy this we produce a dataset of diffusion outputs using COCO [9] as a source of prompts likely to produce images containing





Figure 6. Both sections of ground truth images ((a) & (b)) are generated from the same seed, with (a) utilizing only the positive prompt and (b) incorporating both positive and negative prompts. The images generated using the learned merged prompt consist of two sections ((c) & (d)), where (c) is generated with the same seed as (a) & (b), and (d) is produced using a random seed and learned merged prompt passed through the pre-trained stable diffusion pipeline.

known objects. To look for good prompts we test each image in COCO validation and the first 10% of COCO training split. If  $\exists$  a segmentation  $s$  in the list of segmentations for the image  $i$  s.t.  $area(s) > 30\% * area(i)$  then we add  $i$  to our dataset where the associated prompts  $p^+$  = the first COCO caption, and  $p^-$  = the name of the largest  $s$ . If  $p^- \in p^+$  then it is replaced with a space in  $p^+$  at all occurrences. We do this to remove the case where the negative prompt directly includes a word in the positive. Such instances cause unusual occurrences we do not address in this

work.

Each element of this dataset is then comprised of:  $[COCO\_ID, e^+, e^-, l^+, l^-]$ .  $COCO\_ID$ : the COCO\_ID which produced  $p^{+/-}$ .  $e^{+/-}$ : are the positive and negative embeddings of the prompts respectively.  $l^-$ : the latents produced from each step of diffusion where  $l^-[0]$  = the random sampled model input, and  $l^-[-1]$  = the output of the model.  $l^+$ : the latents produced from  $diffusion(p^+, p^-)$ .  $l^+$ : the latents produced from  $diffusion(p^+)$ .

As there is no guarantee that the produced  $l^+$  when de-

Table 1. Tabular Results of filtering Explorative Diffusion Dataset

| Sr No                                                       | Analysis                                                                                                                                                                                                                                                                                 | Result (5000 prompt set)           |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1 p(pos)                                                    | Percentage of our positive images that contain the “negative prompt word” object                                                                                                                                                                                                         | 18.8%, Datapoints: 940 images      |
| 2 p(Neg)                                                    | Percentage of the negative images that contain the “negative prompt word” object                                                                                                                                                                                                         | 9.61%, Datapoints: 480 images      |
| 3 Conditional Prob: $p(\text{neg} = 1 \mid \text{pos} = 0)$ | Likelihood of the negative image containing the object, given the positive image does.                                                                                                                                                                                                   | 55.21%, Datapoints: 517/940 images |
| 3 Conditional Prob: $p(\text{neg} = 0 \mid \text{pos} = 1)$ | Likelihood of the negative image not containing the object, given the positive image does.                                                                                                                                                                                               | 45%, Datapoints: 216/480 images    |
| 4 Joint prob: $p(\text{neg}, \text{pos})$                   | Calculated as the ratio of data points where both the negative image contains the object (NegHasNeg = 0) and the positive image contains the object (PosHasNeg = 1), to the total data points in the dataset. This represents the likelihood of both conditions being met simultaneously | 10.38%, Datapoints: 519 images     |

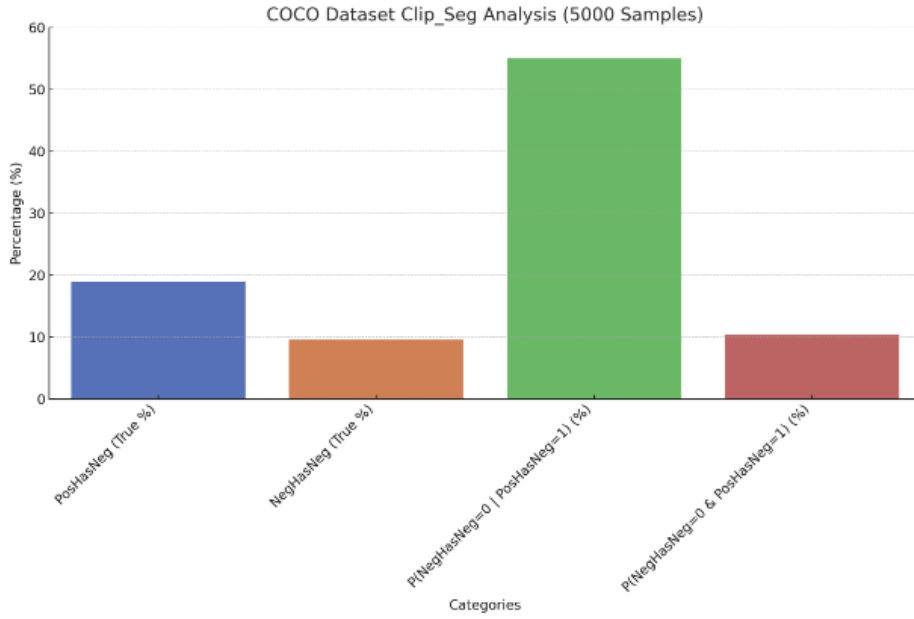


Figure 7. Results of filtering Explorative Diffusion Dataset

coded produces an image that contains  $p^-$ , we perform CLIP-seg in Figure 7 to identify whether the object exists in the image and whether it makes up a sufficiently large portion of the image to be considered ( $> 30\%$ ). These filters on the dataset dilute down our model passes to those where we know the positive prompt doesn’t contain the negative prompt, and we know that the object described in the negative prompt is present when the positive prompt alone produces an image.

We tested the 5,000 images we got from COCO validation which passed our initial prompt filter. Of those, 940 contain the object of interest in the positive prompt, and of

those 55.03% contain the object.

## 5. Related Works

**Impact of Negative Prompts** This study is primarily driven by the imperative to unravel the obscured dimensions of negative prompt influence in image synthesis. The introduction of the Diffusion Model [17] marked a seminal moment in the landscape of generative models, prompting subsequent research endeavors to refine and enhance its capabilities. Noteworthy contributions include DDPM [5], Beating GANs [2], Stable Diffusion [13], DALL-E 2 [12], and Imagen [15]. **Opacities in Model Explainability:** While

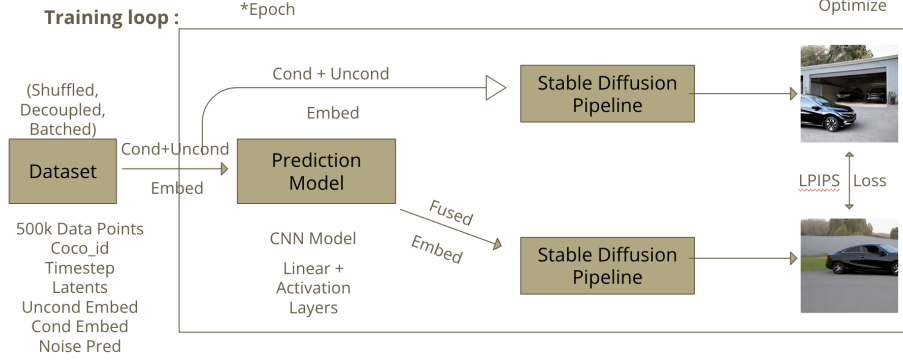


Figure 8. Fused Prompt Prediction Model (FPPM)

the evolution of diffusion models, particularly exemplified by tools like Imagen [15], has facilitated the generation of remarkably realistic images, the elucidation of their internal mechanisms remains a neglected realm. Considerable strides have been made in unraveling the intricacies of image generation tasks within domains such as object detection and segmentation, notably with Vision Transformers (ViTs) [3]. Furthermore, the refinement of predictive noise through Denoising Diffusion Probabilistic Models (DDPM) [5] has garnered attention. However, despite these advancements setting benchmarks in image classification, the interpretability and negative prompt’s role in the context of image generation through diffusion models have been conspicuously overlooked.

**Optimized Prompt Discovery** [20] proposed a gradient-based method to streamline the optimization of challenging prompts in diffusion models, facilitating easy control over the image generation process. While their approach effectively produces semantically similar images by focusing solely on positive prompts, our study takes a step further. We aim to identify a fused prompt by discrete token optimization that combines both positive and negative elements, generating images comparable in semantic likeness to those produced by separate positive and negative prompts. This exploration provides insights into how stable diffusion models utilize a combined prompt, shedding light on their intricate image-generation mechanisms.

**Interpretation Gaps in Understanding Negative Prompts:** The interpretability of diffusion models holds paramount importance for ensuring accountability, transparency, and alignment with human values in AI systems. Existing works, such as those in Medical Imaging [7] and Understanding Diffusion [11], address issues like the truthfulness of predictions and provide theoretical foundations for diffusion models. Despite comprehensive theoretical approaches, these studies fall short of exploring the intricacies and consequences of negative prompts in practical implementations. The omission of the exploration of negative

prompts in papers like Composable Diffusion Models [10] and Contextualizing Diffusion [16] represents a critical gap in the literature, leaving a pivotal area for our investigation.

## 6. Future Works

### 6.1. Fused Prompt Prediction Model (FPPM)

The Fused Prompt Prediction Model (FPPM) is a neural network designed to directly predict fused embeddings from conditional and unconditional embeddings, yielding semantically similar images. The model architecture includes multiple linear and activation layers, and it is trained on the entire dataset, comprising 500,000 data points from 4.3. The model’s architecture is visually depicted in Figure 8.

### 6.2. Enhancing Prompt Quality with BLEURT

Currently, the DTO model (3.2) is not generating meaningful prompts, as illustrated in Figure 6. Integrating BLEURT [18] can significantly improve prompt quality, thereby enhancing our understanding of the impact of negative prompts in a stable diffusion pipeline.

BLEURT offers a quantitative measure of how well the generated text aligns with human-generated reference text. Consequently, it can be employed in conjunction with Mean Squared Error (MSE) as a combined metric for assessing the performance of DTO (3.2). While MSE serves as a distance metric for calculating the loss between learned and merged embeddings, BLEURT specifically evaluates the quality of machine-translated text in comparison to human-reference text.

This work shows two explorations into how negative prompts work in diffusion models. We show that it is possible to optimize over the input tokens to achieve a semantically similar image with a single fused prompt. this allows a kind of probing into what the model is producing novel to diffusion literature. We show that negative prompts frequently do not remove the object in question. Both of these show that the use of the negative prompt should be thought

of as a modifier or even additive process rather than something akin to inpainting.

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